Question - 1:

Normalize the following numbers:

	Given Number	Normalized Number
i.	0.0000124678_{10}	1.24678×10^{-5}
ii.	$1584.234_{10} \times 10^5$	1.584234×10^8
iii.	4782.2354_{10}	4.7822354×10^3
iv.	110101.1111_2	1.101011111×2^5
v.	0.001100_2	1.100×2^{-3}
vi.	$1101.1111_2 \times 2^5$	$1.101\ 1111 \times 2^8$

Question - 2:

Find the Biased Exponent of 1.1011×2^{34} in IEEE-754 single precision format.

(actual exp. — normalized number — Exp = 8 bits)

$$\text{Bias} = 2^{8-1} - 1 = 127$$

$$\text{Biased exponent} = 34 + 127$$
$$= 161$$
$$= 1010\ 0001 \quad (\text{Ans})$$

Question - 3:

Find the Biased Exponent of 1.1011 x 2^{4} in 12-bit IEEE-754 format where the size of the exponent field is 4 bits.

actual exponent

normalized number

$Bias = 2^{4-1} - 1$

$= 7$

Biased exponent = 4 + 7

= 11

= 1011 (Ans)

Question - 4:

Find the Biased Exponent of 1.1011 x 2^{34} in 64-bit IEEE-754 format.

actual exp

normalized number

$Bias = 2^{11-1} - 1$

$= 1023$

Biased Exponent = (34 + 1023) = 1057

= 100 0010 0001 (Ans)

Question - 5:

Find the Biased Exponent of 5678.898 in 34-bit IEEE-754 format where the size of the exponent field is 10 bits.

5678.898

= 1 0110 0010 1110 . 1110 0101 11 . . . .

= $1.0110\ 0010\ 1110\ 1110 \times 2^{12}$ (12 = actual exponent)

$Bias = 2^{10-1} - 1 = 511$

Biased exponent = 511 + 12 = 523

= 10 0000 1011 (Ans)

Question - 6: sign bit = 1

Convert -0.00987_{10} in 34-bit IEEE-754 floating point representation where the size of the fraction field is 23 bits.

sign bit	exponent	fraction
1	10	23

$0.00987 = 0.0000\ 0010\ 10\ldots\ldots$

$= 1.010\ldots. \times 2^{-7}$ ← actual exponent

Fraction

$Bias = 2^{10-1} - 1 = 511$

Biased Exponent = $511 + (-7) = 504 = 01\ 1111\ 1000$

Fraction = $0100\ 0000\ 0000\ 0000\ 0000\ 000$

1	01 1111 1000	0100 0000 0000 0000 0000 000
	2 F C	2 0 0 0 0 0

$= 0x2FC200000$ (Ans)

try Q7 - Q8 yourself

Question - 9:

Convert the following IEEE-754 single-precision floating point numbers into decimal.

	Given Numbers	Decimal Representation
i.	0xFF1205BA	$-1.9409702) \times 10^{38}$
ii.	3457890989_{10}	try yourself
iii.	23245613451_8	

(i) 0xFF1205BA

= 1111 1111 0001 0010 0000 0101 1011 1010

sign Bit | Biased Exp. | Fraction

⇒ Bias = $2^{8-1} - 1 = 127$

Biased exponent = 1111 1110 = 254

∴ Actual q = 254 − 127 = 127

⇒ Fraction = 0.001 0010 0000 0101 1011 1010

= 0·1407 9976 0818 4814 4531

⇒ Decimal value = $(-1)^1 \times (1 + 0.1407\ 9976\ 0818\ 4814\ 4531) \times 2^{127}$

$= -1.1407\ 9976\ 0818\ 4814\ 4531 \times 2^{127}$

$= -1.9409702) \times 10^{38}$ (Ans)

Question - 10:

Multiply the given numbers using IEEE-754 single-precision floating-point
representation. Check if the result has overflow or underflow. Show the result in decimal.

Note: Consider 10 decimal digits while converting from decimal to binary for the following questions.

	Given Numbers	Result	Overflow/ Underflow
i.	7.01_{10} and 0.71_{10}		
ii.	0.000101_2 and 10.1_2		
iii.	$0.000101_2 \times 2^{70}$ and $10010.000101_2 \times 2^{60}$		
iv.	1584.234_{10} and 1584.234_{10}		
v.	0.001100_2		
vi.	$1101.1111_2 \times 2^5$ and $110.000101_2 \times 2^6$		

(iii) $0.000101 \times 2^{70} \times 10010.000101 \times 2^{60}$

$\Rightarrow 0.000101 \times 2^{70}$

$= 1.01 \times 2^{70-4}$

$= 1.01 \times 2^{66}$

$\Rightarrow 10010.000101 \times 2^{60}$

$\Rightarrow 1.0010\ 0001\ 01 \times 2^{60+4}$

$= 1.0010\ 0001\ 01 \times 2^{64}$

$\Rightarrow 1.01 \times 2^{66} \times 1.0010\ 0001\ 01 \times 2^{64}$

$\Rightarrow (1.01 \times 1.0010\ 0001\ 01) \times 2^{130}$

$\Rightarrow 1.011101 \times 2^{130}$ ← actual exponent

Bias $= 2^{8-1} - 1 = 127$

Biased Exponent $= 127 + 130$

$= 257$

Range of Biased exponent: 0 to $2^8 - 1$

$= 0$ to 255

$= (1$ to $254)$ [0 & 255 reserved]

(upper range) 254 is smaller than biased exponent (257)

Hence, Overflow.

(iv) 1584.234 × 1584.234

1584 = 110 0011 0000

.234 = .0011 1011 11....

1584.234 = 110 0011 0000. 0011 1011 11....

$= 1.10\ 0011\ 0000\ 0011\ 1011\ 11.... \times 2^{10}$

$\Rightarrow 1.10\ 0011\ 0000\ 0011\ 1011\ 11.... \times 2^{10} \times 1.10\ 0011\ 0000\ 0011\ 1011\ 11.... \times 2^{10}$

$= 10.01100 \times 2^{20}$

$= 1.0011\ 00 \times 2^{21}$

Bias $= 2^{8-1} - 1 = 127$

Biased Exponent $= 127 + 21$

$= 148$

Range of Biased exponent: 0 to $2^8 - 1$

$= 0$ to 255

$= (1$ to $254)$ [0 & 255 reserved]

$1 < 148 < 254$

Hence, No overflow / underflow

Question - 11:

Multiply the given numbers using IEEE-754 double-precision floating-point
representation. Check if the result has overflow or underflow.

Note: Consider 10 decimal digits while converting from decimal to binary for the following questions.

	Given Numbers	Result	Overflow/ Underflow
i.	7.01_{10} and 0.71_{10}	Try next yourself	
ii.	$0.000101_2 \times 2^{-850}$ and $10.1_2 \times 2^{-900}$		
iii.	$0.0101_2 \times 2^{790}$ and $10010.0101_2 \times 2^{680}$		

(iii) $0.0101 \times 2^{790} \times 10010.0101 \times 2^{680}$

$\Rightarrow 0.0101 \times 2^{790}$

$= 1.01 \times 2^{790-2}$

$= 1.01 \times 2^{788}$

$\Rightarrow 10010.0101 \times 2^{680}$

$= 1.0010\,0101 \times 2^{680+4}$

$= 1.0010\,0101 \times 2^{684}$

$\Rightarrow 1.01 \times 2^{788} \times 1.0010\,0101 \times 2^{684}$

$= (1.01 \times 1.0010\,0101) \times 2^{788+684}$

$= 1.01101 \times 2^{1472}$ ← actual exponent.

Bias $= 2^{11-1} - 1 = 1023$

Biased Exponent $= 1023 + 1472$

$= 2495$

Range of Biased exponent: 0 to $2^{11}-1$

$= 0$ to 2048

$= (1$ to $2047)$ [0 & 2048 reserved]

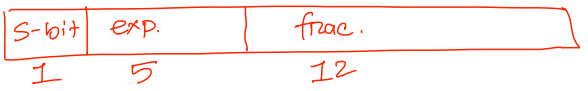
$2495 > 2047$ (upper range)

Hence, Overflow.

Question - 12:

Multiply the given numbers using 18bit IEEE-754 floating-point representation where the size of the fraction field is 12 bits. Check if the result has overflow or underflow.

Note: Consider 10 decimal digits while converting from decimal to binary for the following questions.



	Given Numbers	Result	Overflow/ Underflow
i.	7.01_{10} and 0.71_{10}		
ii.	$0.000101_2 \times 2^{-85}$ and $10.1_2 \times 2^{-90}$		
iii.	$0.0101_2 \times 2^{79}$ and $10010.0101_2 \times 2^{68}$		

(ii) $0.000101 \times 2^{-85} \times 10.1 \times 2^{-90}$

$\Rightarrow 0.000101 \times 2^{-85}$

$= 1.01 \times 2^{-85-4}$

$= 1.01 \times 2^{-89}$

$\Rightarrow 10.1 \times 2^{-90}$

$= 1.01 \times 2^{-90+1}$

$= 1.01 \times 2^{-89}$

$\Rightarrow 1.01 \times 2^{-89} \times 1.01 \times 2^{-89}$

$= (1.01 \times 1.01) \times 2^{-89-89}$

$= 1.1001 \times 2^{-178}$ — actual exponent

Bias $= 2^{5-1} - 1 = 15$

Biased Exponent $= 15 - 178$

$= -163$

Range of Biased exponent: 0 to $2^5 - 1$

$= 0$ to 31

$= (1$ to $30)$ [0 & 31 reserved]

$-163 < 1$ (lower range)

Hence, underflow.